

A HIV

Noise σ	Bootstrap	Column-wise	GMM	SMOTE	GMM-SMOTE	CVAE
0	0.721 $\pm$ 0.066	0.900 $\pm$ 0.044	1.000 $\pm$ 0.000	0.577 $\pm$ 0.032	0.570 $\pm$ 0.042	1.000 $\pm$ 0.000
0.05	0.837 $\pm$ 0.047	0.999 $\pm$ 0.001	1.000 $\pm$ 0.000	0.823 $\pm$ 0.046	0.812 $\pm$ 0.069	1.000 $\pm$ 0.001
0.1	0.906 $\pm$ 0.038	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.886 $\pm$ 0.023	0.877 $\pm$ 0.058	1.000 $\pm$ 0.000
0.2	0.980 $\pm$ 0.019	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.985 $\pm$ 0.013	0.954 $\pm$ 0.034	0.999 $\pm$ 0.001
0.3	0.996 $\pm$ 0.004	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.999 $\pm$ 0.002	0.986 $\pm$ 0.013	0.999 $\pm$ 0.002
0.5	0.999 $\pm$ 0.002	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.997 $\pm$ 0.004	0.999 $\pm$ 0.002
0.75	0.999 $\pm$ 0.002	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.999 $\pm$ 0.002
1	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
1.5	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
2	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000

B Breast Cancer

Noise σ	Bootstrap	Column-wise	GMM	SMOTE	GMM-SMOTE	CVAE
0	0.704 $\pm$ 0.023	0.999 $\pm$ 0.000	0.994 $\pm$ 0.002	0.545 $\pm$ 0.015	0.583 $\pm$ 0.023	0.771 $\pm$ 0.029
0.05	0.753 $\pm$ 0.020	0.999 $\pm$ 0.000	0.994 $\pm$ 0.002	0.547 $\pm$ 0.010	0.583 $\pm$ 0.028	0.754 $\pm$ 0.031
0.1	0.678 $\pm$ 0.027	0.999 $\pm$ 0.000	0.994 $\pm$ 0.002	0.530 $\pm$ 0.019	0.568 $\pm$ 0.028	0.743 $\pm$ 0.024
0.2	0.683 $\pm$ 0.026	0.998 $\pm$ 0.001	0.995 $\pm$ 0.002	0.679 $\pm$ 0.022	0.650 $\pm$ 0.028	0.805 $\pm$ 0.012
0.3	0.869 $\pm$ 0.018	0.996 $\pm$ 0.001	0.995 $\pm$ 0.002	0.862 $\pm$ 0.011	0.844 $\pm$ 0.022	0.886 $\pm$ 0.010
0.5	0.966 $\pm$ 0.010	0.996 $\pm$ 0.003	0.997 $\pm$ 0.002	0.965 $\pm$ 0.006	0.962 $\pm$ 0.008	0.972 $\pm$ 0.004
0.75	0.991 $\pm$ 0.003	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.989 $\pm$ 0.004	0.990 $\pm$ 0.003	0.992 $\pm$ 0.003
1	0.997 $\pm$ 0.002	0.999 $\pm$ 0.001	0.999 $\pm$ 0.001	0.996 $\pm$ 0.002	0.996 $\pm$ 0.002	0.998 $\pm$ 0.001
1.5	0.999 $\pm$ 0.000	0.999 $\pm$ 0.001	0.999 $\pm$ 0.000	0.999 $\pm$ 0.001	0.999 $\pm$ 0.001	0.999 $\pm$ 0.001
2	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000

C Diabetes

Noise σ	Bootstrap	Column-wise	GMM	SMOTE	GMM-SMOTE	CVAE
0	0.719 $\pm$ 0.019	0.777 $\pm$ 0.016	0.979 $\pm$ 0.007	0.712 $\pm$ 0.021	0.714 $\pm$ 0.020	0.979 $\pm$ 0.003
0.05	0.938 $\pm$ 0.009	0.982 $\pm$ 0.005	0.981 $\pm$ 0.007	0.963 $\pm$ 0.006	0.965 $\pm$ 0.006	0.978 $\pm$ 0.004
0.1	0.945 $\pm$ 0.010	0.981 $\pm$ 0.004	0.981 $\pm$ 0.005	0.962 $\pm$ 0.008	0.963 $\pm$ 0.007	0.975 $\pm$ 0.004
0.2	0.954 $\pm$ 0.009	0.982 $\pm$ 0.005	0.978 $\pm$ 0.005	0.966 $\pm$ 0.010	0.966 $\pm$ 0.008	0.970 $\pm$ 0.001
0.3	0.965 $\pm$ 0.008	0.984 $\pm$ 0.003	0.978 $\pm$ 0.005	0.969 $\pm$ 0.007	0.967 $\pm$ 0.007	0.968 $\pm$ 0.004
0.5	0.974 $\pm$ 0.006	0.985 $\pm$ 0.004	0.978 $\pm$ 0.004	0.975 $\pm$ 0.006	0.975 $\pm$ 0.004	0.973 $\pm$ 0.006
0.75	0.981 $\pm$ 0.007	0.989 $\pm$ 0.001	0.984 $\pm$ 0.006	0.984 $\pm$ 0.005	0.983 $\pm$ 0.004	0.982 $\pm$ 0.003
1	0.986 $\pm$ 0.005	0.989 $\pm$ 0.003	0.990 $\pm$ 0.004	0.990 $\pm$ 0.002	0.989 $\pm$ 0.001	0.988 $\pm$ 0.005
1.5	0.993 $\pm$ 0.001	0.994 $\pm$ 0.002	0.994 $\pm$ 0.001	0.996 $\pm$ 0.001	0.995 $\pm$ 0.001	0.992 $\pm$ 0.002
2	0.996 $\pm$ 0.001	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.996 $\pm$ 0.001